Achieving Optimal Fraud Detection with PaySim: A Comparative Study of Random Forest and Decision Tree

Final Examination

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Source Code: GitHub Repository

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Executive Summary

The rapid digitalization of financial transactions necessitates robust mechanisms for detecting anomalous and fraudulent behavior. This project details the analysis and modeling of the PaySim synthetic financial dataset, containing over 6.3 million transaction records. Due to an extreme class imbalance (less than 0.14% fraud), a systematic preprocessing pipeline was developed, utilizing One-Hot Encoding, StandardScaler normalization, and Random Undersampling. A stratified train/test split was performed *before* undersampling to ensure the test set preserved the real-world class distribution. Two classification models—Random Forest and Decision Tree—were then trained on the balanced training subset and evaluated against the realistic imbalanced test set. A direct comparison of both models revealed that Random Forest outperformed the single Decision Tree across all metrics, achieving a 99.6% recall rate versus 99.1%, a higher ROC-AUC of 0.9995 versus 0.9922, and a superior F1-score, confirming that ensemble methods provide measurable gains in fraud detection performance.

1 Introduction and Exploratory Data Analysis (EDA)

The primary obstacle in financial fraud detection is *class imbalance*; standard algorithms tend to blindly predict the majority class (legitimate transactions) to achieve artificially high accuracy, completely missing the rare fraudulent events. Initial analysis of the 6.3 million rows revealed that fraudulent transactions were exclusively isolated to `TRANSFER` and `CASH_OUT` types and exhibited significantly higher median monetary values and volatile bursts compared to normal transactions, which were tightly clustered at lower dollar amounts.

Figure 1: Original Class Distribution. Note the logarithmic scale required to visualize the rare fraud cases.

Figure 2: Transaction Type vs. Fraud. Fraud is exclusively concentrated in TRANSFER and CASH_OUT types.

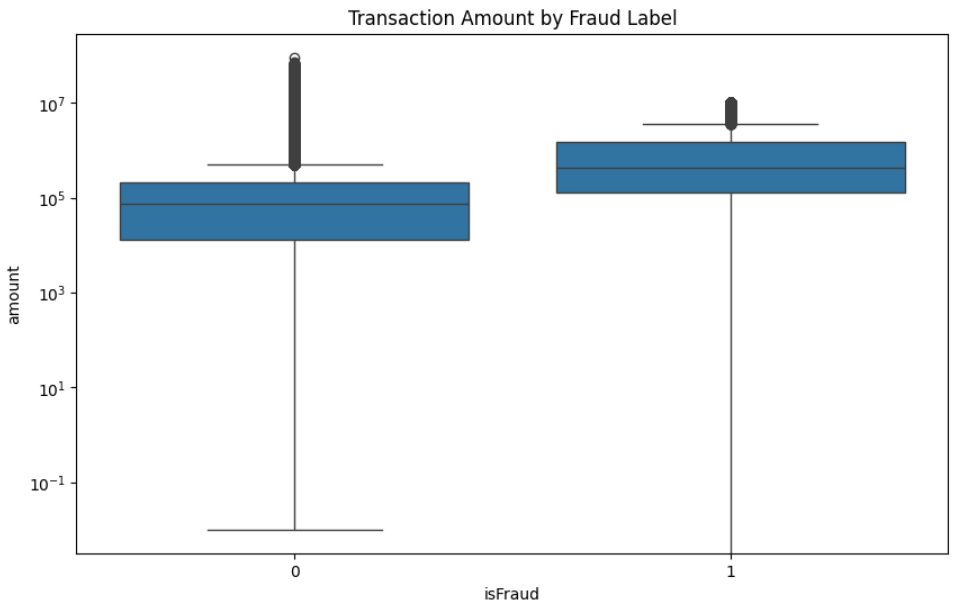

Figure 3: Transaction Amount Distribution. Fraudulent transactions exhibit significantly higher and more volatile monetary values compared to legitimate transactions.

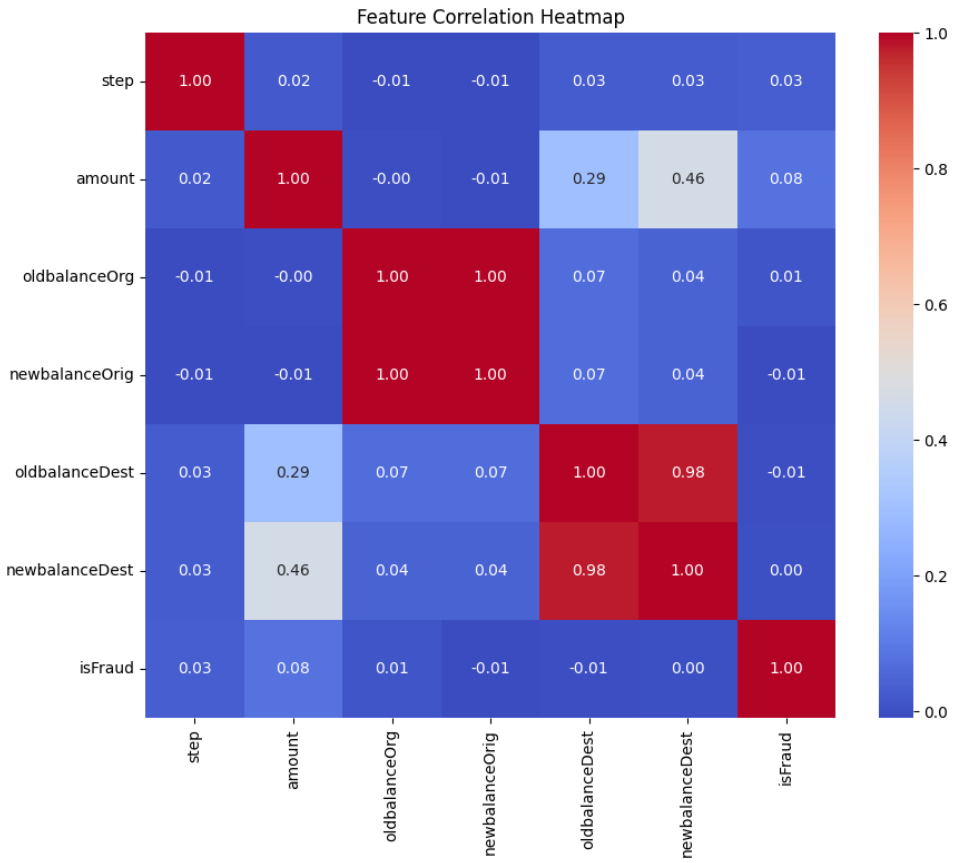

Figure 4: Feature Correlation Heatmap. Strong negative correlations between transaction amounts and sender balances suggest that fraudulent transfers tend to drain accounts.

2 Methodology and Data Preprocessing

To prepare the data for the classification models, a multi-step transformation pipeline was executed. Critically, the dataset was split into training and testing sets *before* any balancing was applied, ensuring the evaluation reflects real-world performance.

- **Data Cleaning:** Identifier columns (`nameOrig`, `nameDest`) and the redundant `isFlaggedFraud` column were removed, as they carry no predictive signal and would cause overfitting or data leakage if retained.
- **Categorical Encoding:** The `type` column was One-Hot Encoded (with `drop_first=True` to avoid multicollinearity) to allow independent mathematical comparison by the model.
- **Stratified Train/Test Split:** The full dataset was split 80/20 using `stratify=y`, which preserves the original 0.13% fraud rate in both the training and testing sets. This ensures the test set reflects real-world conditions.
- **Feature Scaling (Standardization):** We applied Z-score normalization (StandardScaler) to the numeric features. The scaler was fit exclusively on the training set and then used to transform both sets, preventing data leakage. Note: while

tree-based models are inherently scale-invariant (they split on thresholds, not distances), scaling was applied as a best practice for pipeline consistency and potential future model comparisons with distance-based algorithms.

- **Data Balancing (Random Undersampling):** To resolve the extreme class imbalance, undersampling was applied *only to the training set*. Normal transactions were randomly downsampled to match the number of fraudulent transactions, creating a balanced 50/50 training subset. The test set was left untouched at its original imbalanced distribution to ensure realistic evaluation.

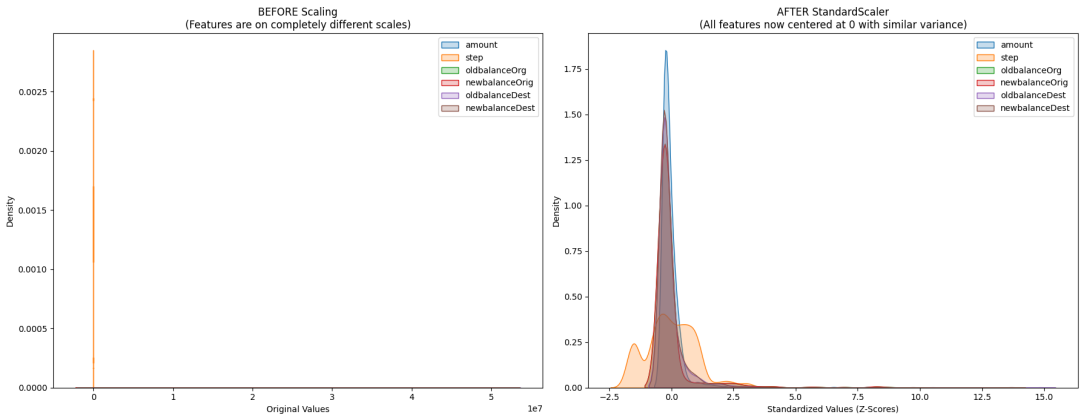

Figure 5: Feature Scaling Proof. KDE plot comparing numeric features before and after applying StandardScaler.

3 Model Training and Evaluation

Two classification models were trained on the balanced training subset and evaluated against the full imbalanced test set of approximately 1,272,524 transactions (roughly 1,270,881 legitimate and 1,643 fraudulent). Both models share the same preprocessing pipeline, train/test split, and balanced training data, ensuring a fair and controlled comparison.

3.1 Model 1: Random Forest Classifier

The Random Forest Classifier was configured with 100 decision trees (`n_estimators=100`). Random Forest is an *ensemble* method that constructs multiple independent decision trees and aggregates their predictions via majority voting. This averaging mechanism reduces variance and overfitting compared to any single tree.

The model achieved a **Recall of 99.6%** for the fraud class (1,637 out of 1,643 actual fraud cases detected), with only 6 fraudulent transactions missed. The realistic evaluation also revealed the expected **precision–recall tradeoff**: the model flagged 16,181 legitimate transactions as suspicious (false positives), yielding a fraud precision of approximately 9.2%. In a production environment, these flagged transactions would be routed to a human review queue—a standard practice where the cost of missing real fraud far outweighs the cost of investigating false alarms.

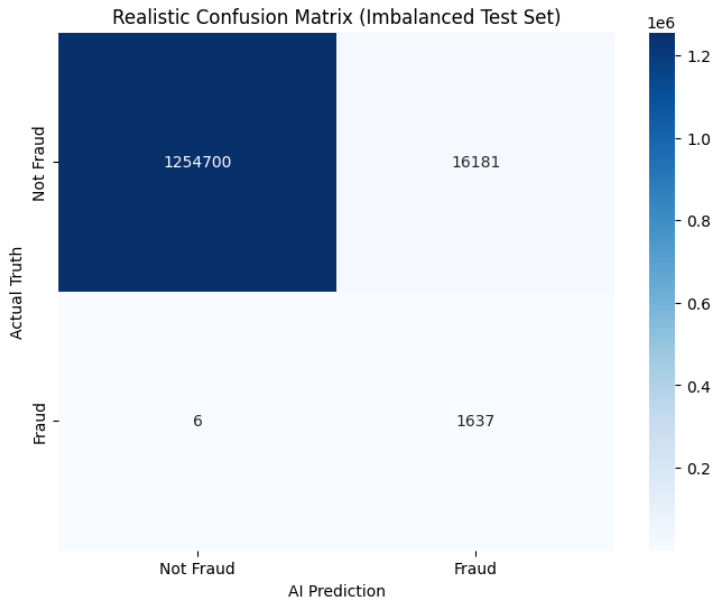

Figure 6: Random Forest Confusion Matrix (Imbalanced Test Set). The model correctly identified 1,637 out of 1,643 actual fraud cases, while flagging 16,181 legitimate transactions for review.

3.2 Model 2: Decision Tree Classifier

A single Decision Tree Classifier was trained with a maximum depth of 10 (`max_depth=10`) to provide a controlled comparison against the ensemble approach. The Decision Tree is the fundamental building block of Random Forest; comparing the two directly demonstrates the quantitative benefit of ensembling.

The Decision Tree achieved a **Recall of 99.1%** for the fraud class (1,629 out of 1,643 fraud cases detected), missing 14 fraudulent transactions—more than double the 6 missed by Random Forest. It also produced 18,686 false positives (compared to 16,181 for RF), yielding a lower fraud precision of approximately 8.0%.

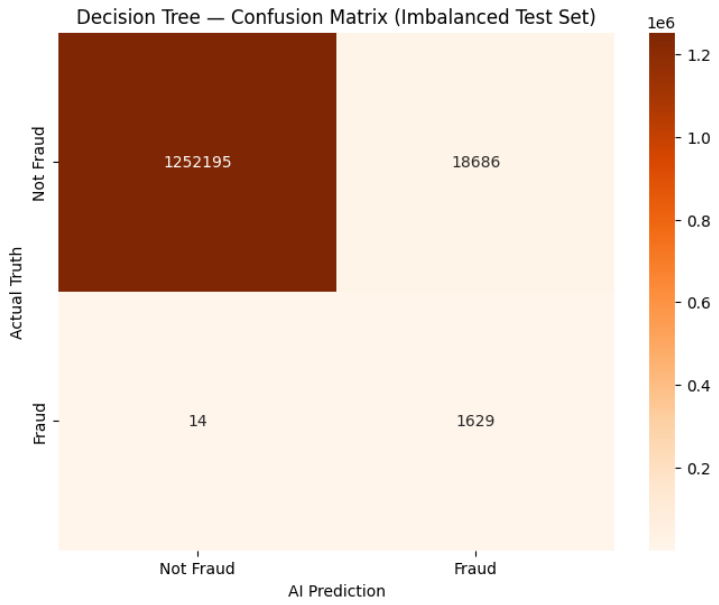

Figure 7: Decision Tree Confusion Matrix (Imbalanced Test Set). The single tree missed 14 fraud cases and generated more false positives than the Random Forest.

3.3 Model Comparison

A side-by-side comparison of all key metrics confirms that Random Forest consistently outperforms the single Decision Tree across every evaluation dimension.

Model	**Accuracy**	**Prec. (Fraud)**	**Recall (Fraud)**	**F1 (Fraud)**	**ROC-AUC**
Random Forest	0.9873	0.0919	0.9963	0.1683	0.9995
Decision Tree	0.9853	0.0802	0.9915	0.1484	0.9922

Table 1: Performance comparison of Random Forest vs. Decision Tree on the imbalanced test set. Random Forest leads in all metrics.

The ROC curve overlay provides the clearest visualization of this gap: Random Forest's curve hugs the top-left corner more tightly (AUC = 0.9995) than the Decision Tree (AUC = 0.9922), indicating superior discrimination between fraud and non-fraud at every classification threshold.

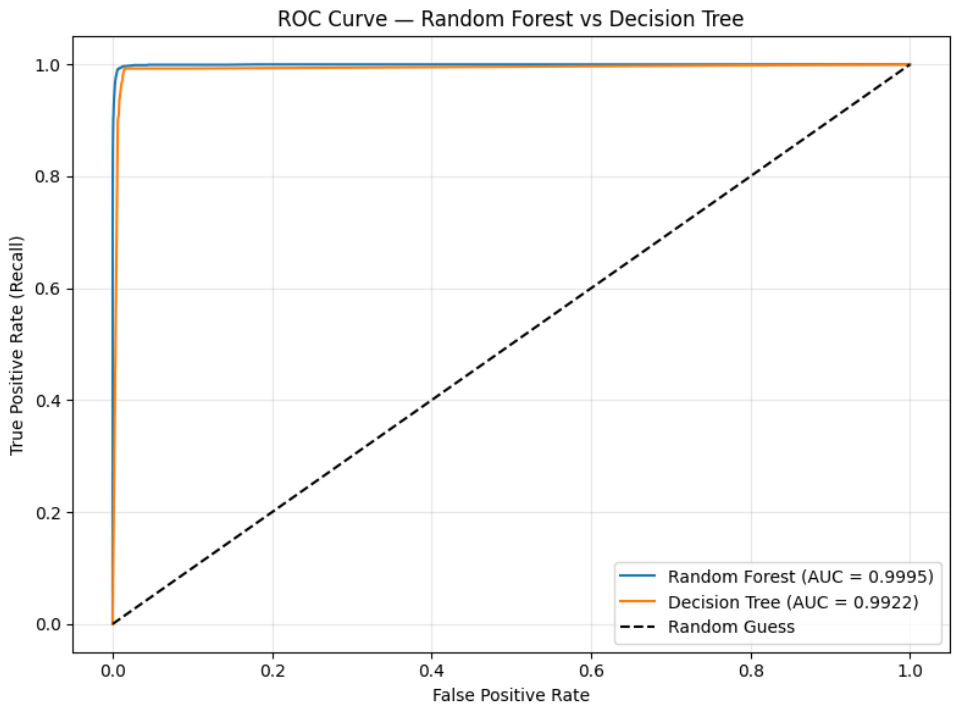

Figure 8: ROC Curve Comparison. Random Forest (AUC = 0.9995) consistently outperforms the Decision Tree (AUC = 0.9922) across all threshold settings.

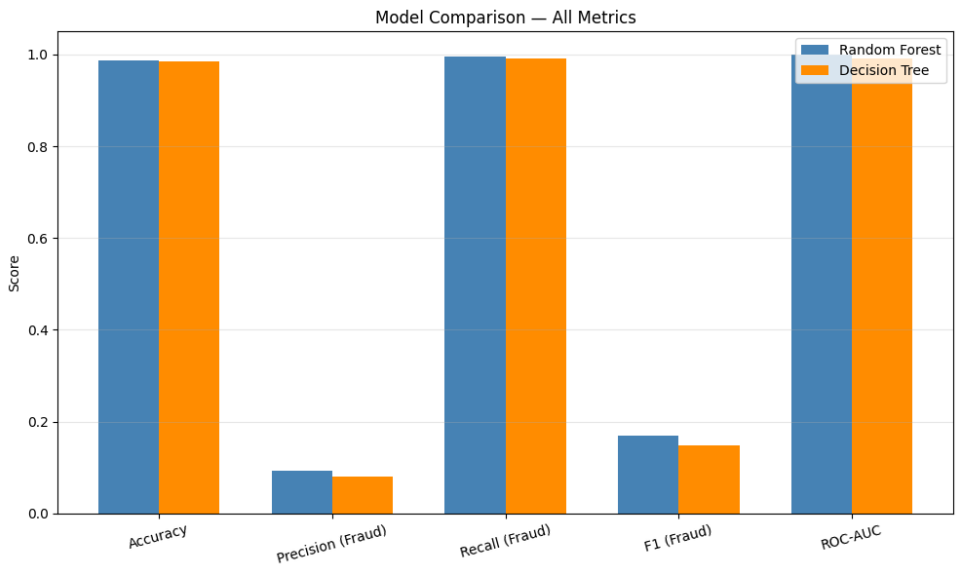

Figure 9: Side-by-side bar chart of all evaluation metrics. Random Forest (blue) leads in every category.

The reason for this performance gap is structural: a single Decision Tree is prone to high variance—small changes in the training data can produce a very different tree. Random Forest mitigates this by training 100 independent trees on bootstrapped samples and averaging their votes, which smooths out individual errors and produces more stable, generalizable predictions.

3.4 Feature Importance

The Random Forest model relied most heavily on variables related to sender and receiver account balances rather than just the raw transaction `amount`. This aligns with the known fraud pattern in the dataset: fraudulent transfers tend to drain sender accounts and inflate receiver accounts, making the balance delta columns the strongest predictive signals.

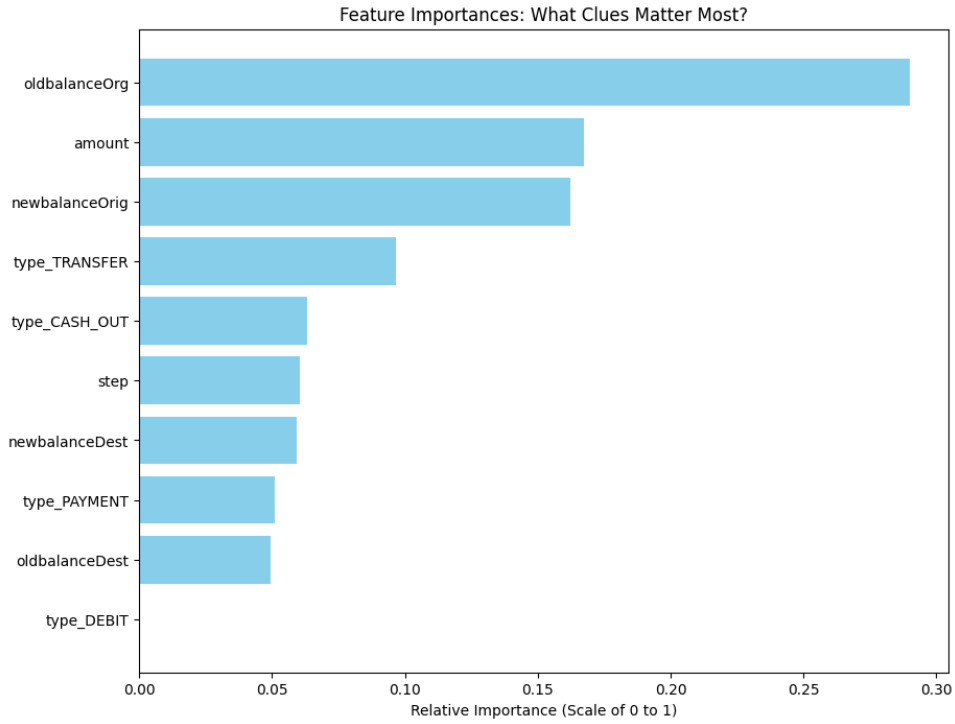

Figure 10: Feature Importance Ranking. Account balance features dominate, confirming that balance anomalies are the strongest fraud indicators.

4 Key Findings and Insights

- **Ensemble methods outperform single models:** Random Forest achieved a 0.73% higher ROC-AUC and caught 8 additional fraud cases compared to the Decision Tree, demonstrating that aggregating multiple weak learners produces measurably stronger predictions.

- **Recall is the critical metric:** In fraud detection, each missed case represents direct financial loss. The Random Forest's 99.6% recall means only 6 out of 1,643 fraudulent transactions were missed across the entire test set.

- **The precision–recall tradeoff is manageable:** Both models produced low fraud precision (9% and 8%), which is expected and acceptable in production systems where flagged transactions are routed to human reviewers rather than automatically blocked.

- **Balance features are the strongest fraud signals:** Feature importance analysis revealed that sender and receiver account balances—not raw transaction amounts—

are the most predictive features, consistent with the fraud pattern of draining accounts via large transfers.

- **Fraud is type-specific:** Exploratory analysis confirmed that fraud occurs exclusively in TRANSFER and CASH_OUT transaction types, which could be leveraged as a rule-based pre-filter in production.

5 Limitations and Future Work

- **Synthetic data:** The PaySim dataset is a simulation, and real-world fraud patterns may be more complex, adversarial, and evolving over time. Model performance on production data would likely differ.
- **Low precision:** Both models produced a high number of false positives. Future work could explore techniques such as SMOTE oversampling, cost-sensitive learning, or threshold tuning to improve precision without sacrificing recall.
- **Limited model exploration:** Only tree-based models were evaluated. Gradient Boosted Trees (XGBoost, LightGBM) and neural network approaches could yield better precision–recall balance and should be benchmarked in future iterations.
- **No temporal validation:** The current evaluation uses a random stratified split. In production, fraud detection models should be validated using a time-based split (train on past, test on future) to account for concept drift.
- **Feature engineering:** Additional derived features—such as transaction velocity (transactions per hour per account), balance change ratios, and historical account behavior—could significantly improve model discrimination.

6 Conclusion

By establishing a robust preprocessing pipeline—specifically leveraging a stratified split before undersampling, proper scaler isolation, and training-only balancing—this project successfully transformed a massive, highly imbalanced dataset into an actionable intelligence tool. The comparative evaluation of Random Forest and Decision Tree confirmed that ensemble methods provide consistent, measurable improvements over single-tree classifiers. The Random Forest model's 99.6% recall rate and near-perfect ROC-AUC of 0.9995 demonstrate that advanced machine learning can effectively isolate and identify complex fraud vectors, while the side-by-side comparison with the Decision Tree provides clear empirical evidence for why ensembling is the preferred approach in production fraud detection systems.